

Deep Learning for Multiphasic CT-Based Liver Cancer Subtype Classification and Segmentation Refinement

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Abstract—Primary liver cancer, including hepatocellular carcinoma (HCC), intrahepatic cholangiocarcinoma (ICC), and combined hepatocellular cholangiocarcinoma (cHCC-iCCA), remains difficult to distinguish on multiphasic contrast-enhanced CT (CECT). Overlapping enhancement patterns make reliable subtype differentiation and accurate tumor boundary delineation challenging. This study proposes a dual-model deep learning framework for multiphasic CECT. HCC-Net uses an EfficientNet-based phase-stacked architecture trained on 278 patients. HCC-RefineNet applies a 3D U-Net within tumor-centered regions of interest. HCC-Net achieved a macro-area under the receiver operating characteristic (ROC) curve (AUC) of 0.98, accuracy of 0.96, and macro-F1 score of 0.96, with balanced per-class performance. HCC-RefineNet achieved a mean Dice of 0.93 (SD 0.05) and HD95 of 2.99 (SD 0.78) within tumor-centered regions of interest. Both models operate on publicly available multiphasic CECT data without additional clinical inputs, establishing a reproducible baseline for three-class primary liver cancer analysis.

Index Terms—Liver cancer, CT imaging, Classification, Segmentation, EfficientNet, U-Net

I. INTRODUCTION

Liver cancer is the sixth most diagnosed cancer worldwide [1], with hepatocellular carcinoma (HCC) comprising approximately 70% of cases, and intrahepatic cholangiocarcinoma (ICC) accounting for 12%. Combined hepatocellular cholangiocarcinoma (cHCC-iCCA), a rare variant exhibiting features of both tumor types, represents about 2-3% of cases [2]. Delayed or missed diagnosis of ICC and cHCC-iCCA often results in loss of the optimal treatment window and poorer patient outcomes, underscoring the urgency for accurate and timely diagnosis [3]. Despite its importance for treatment planning and post-interventional assessment, liver tumor segmentation remains challenging due to anatomical variability and inconsistent contrast-enhanced appearances across phases [4], [5]. This study addresses multiphasic CT-based liver cancer subtype classification and post-hoc tumor boundary refinement. Multiphasic CECT acquires four phases including plain (P), arterial (C1), venous (C2), and delayed (C3) - each capturing distinct enhancement characteristics valuable for subtype discrimination. However, existing frameworks are limited to binary or tri-class settings excluding cHCC-iCCA [6], [7], or rely on additional clinical data to achieve diagnostic performance [8]. Manual tumor masks require expert consensus due to boundary

ambiguity [9], highlighting the need for automated boundary refinement. This paper proposes two complementary models for multiphasic liver cancer analysis: a phase-aware subtype classifier and a post-hoc tumor boundary refinement network. The main contributions are as follows.

- 1) Annotation-free 2D middle-slice baseline for four-phase CECT classification.
- 2) Phase-aligned early-fusion representation stacking four phases (C1, C2, C3, P) as channels for compact subtype discrimination.
- 3) Three-phase tumor boundary refinement using C1, C2, and C3 with an initial mask to produce a cleaner segmentation.
- 4) Reproducible experiments on a publicly available multiphasic CECT dataset, providing a benchmark for future work.

II. MATERIALS AND METHODS

A. Dataset

The primary dataset (ScienceDB [10]) contains 278 patients with histopathologically confirmed liver cancer: HCC (n=94), ICC (n=99), cHCC-iCCA (n=85)¹ and 83 non-cancer controls (CN), all with four-phase CECT. All patients underwent standardized four-phase CECT (P, C1, C2, C3) at a single center, yielding 1,444 NIFTI volumes. Only cases meeting both criteria were included:

- 1) Possession of a confirmed primary liver cancer subtype label (HCC, ICC, or cHCC-iCCA), verified through histopathology or multidisciplinary clinical review.
- 2) Availability of radiologist-validated tumor segmentation masks, which were required only for supervising the HCC-RefineNet segmentation module.

CN cases were excluded; only the three cancer subtypes were used. Demographic metadata and liver masks were excluded. Raw CT images served classification; radiologist-validated masks served as reference labels for HCC-RefineNet [9]. The dataset is fully de-identified and publicly available. External evaluation used the publicly available WAW-TACE dataset

¹Labeled as CHCC in the original dataset; the WHO-recommended term cHCC-iCCA is used throughout.

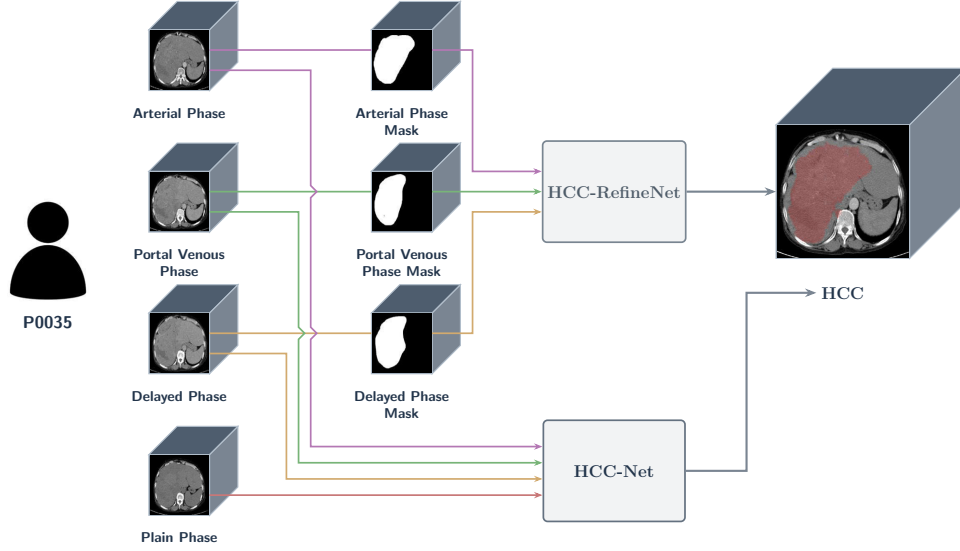


Fig. 1: Overview of the proposed multiphasic CECT pipeline for subtype classification and tumor boundary refinement.

(Zenodo, 233 HCC patients) [11], using CECT phases only, enabling independent assessment of cross-center generalization.

B. Problem formulation

Let $X_{i,j}$ denote multiphasic CECT scans for patient index ($1 \leq i \leq 278$) and $j \in \{P, C1, C2, C3\}$. Two objectives are addressed: (1) classify each case into one of the primary liver cancer subtypes $y_i \in \{\text{HCC}, \text{ICC}, \text{cHCC-iCCA}\}$, and (2) refine tumor segmentation masks for accurate boundary delineation. The two models are applied independently to the same study, as illustrated in Figure 1.

C. Preprocessing

For HCC-Net, each CECT volume was stored in NIfTI format with dimensions $512 \times 512 \times D_{i,j}$, where $D_{i,j}$ ranged from 20 to 347 slices. Singleton dimensions were removed to collapse volumes to 3D. Missing or unreadable phases were handled by duplication or zero-padding to preserve a consistent four-phase input structure. The central axial slice $\text{mid} = \lfloor D_{i,j}/2 \rfloor$ was extracted from each phase as a deterministic, annotation-free representation. Although the tumor center rarely aligns with mid (median offset: 10.5 slices), 60% of patients have a tumor-containing slice within ± 1 slice and 74% within ± 5 slices of mid. Intensities were clipped to a liver-specific window (center 150 HU, width 300 HU; range 0–300 HU) and linearly scaled to $[0, 1]$ to preserve parenchymal and vascular contrast while suppressing high-density noise. All images were resampled to 384×384 using area-based interpolation. The normalized slices from C1, C2, C3, and P were concatenated along the channel dimension to form a $4 \times 384 \times 384$ input tensor fed into an EfficientNet-V2-S backbone [12].

For HCC-RefineNet, C1 served as the spatial reference for multi-phase alignment. Each CECT phase was reshaped by removing singleton axes, converting volumes from $512 \times 512 \times$

$D_{i,j} \times 1 \times 3$ to $512 \times 512 \times D_{i,j}$. Mask files underwent the same standardization and were binarized such that all voxel values greater than zero were set to one. Spatial orientation was standardized to Right-Anterior-Superior (RAS), resulting in $1 \times D_{i,j} \times 512 \times 512$. Phases were resampled to isotropic 1.5 mm spacing. As CECT phases may exhibit minor misalignments due to respiratory motion or scanner offsets, C2 and C3 were resampled using bilinear interpolation, while masks were resampled with nearest-neighbor interpolation to preserve binary integrity, enforcing exact voxel-to-voxel correspondence with the C1 grid. Missing phases were substituted with zero-filled arrays of identical dimensions to C1. Each mask volume $M_{C1}, M_{C2}, M_{C3} \in \{0, 1\}^{D_i \times 512 \times 512}$ was binarized, and a unified ground-truth mask was derived as:

$$L_i = M_{C1} \vee M_{C2} \vee M_{C3}, \quad L_i \in \{0, 1\}^{D_i \times 512 \times 512},$$

ensuring all tumor regions visible in any phase are preserved. The ROI was dilated using a $5 \times 5 \times 5$ structuring element. Center cropping and symmetric padding enforced consistent spatial dimensions. The ROI mask was applied as:

$$\text{img}'_{C1} = \text{img}_{C1} \odot \text{ROI}, \quad \text{lab}' = \text{lab} \odot \text{ROI},$$

retaining only voxels within the ROI. Each phase was windowed at three HU ranges: w1 $[-100, 400]$ emphasizes soft-tissue contrast between tumor and normal liver; w2 $[-160, 240]$ enhances subtle differences between healthy and diseased parenchyma; w3 $[0, 200]$ highlights hyperenhancing regions in CECT. Each windowed image was Gaussian-smoothed, and all windowed and smoothed images from the three phases were concatenated, producing:

$$X_i \in \mathbb{R}^{18 \times D_i \times 512 \times 512},$$

where 18 represents three HU windows and their smoothed versions for each of the three phases.

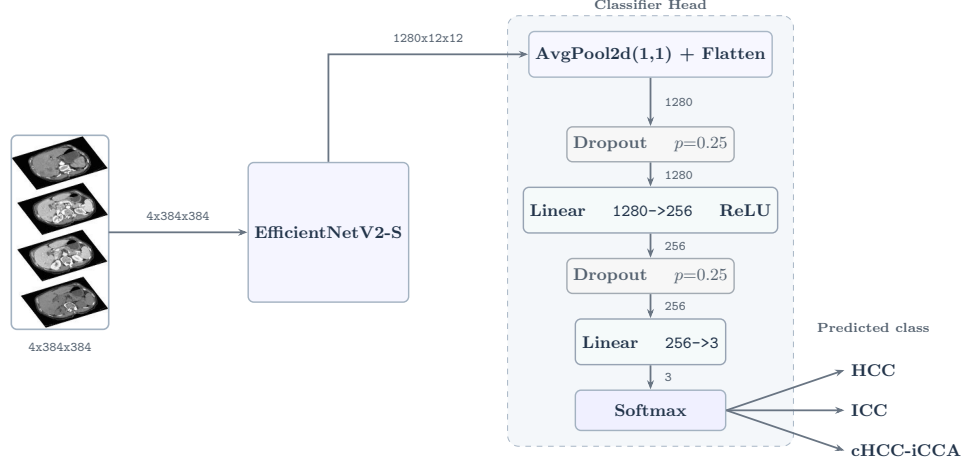


Fig. 2: Architecture of HCC-Net for multiphasic liver cancer subtype classification.

D. Model Architecture

HCC-Net takes a $4 \times 384 \times 384$ phase-stacked tensor as input. An EfficientNet-V2-S backbone [12] extracts a $1280 \times 12 \times 12$ feature map, reduced to $\mathbb{R}^{1280}$ via global average pooling. A two-layer MLP ($1280 \rightarrow 256 \rightarrow 3$) with ReLU and softmax produces class probabilities over $\{\text{HCC}, \text{ICC}, \text{cHCC-iCCA}\}$. The detailed architecture of HCC-Net is illustrated in Figure 2.

HCC-RefineNet is a 3D U-Net [13] taking $X_i \in \mathbb{R}^{18 \times D_i \times 512 \times 512}$ as input. The encoder consists of five stages with channel widths (32, 64, 128, 256, 512) and stride-2 down-sampling, each with two residual convolutional blocks. Skip connections fuse encoder and decoder features at each scale. A final $1 \times 1 \times 1$ convolution followed by sigmoid produces voxel-wise probabilities; binary masks are obtained by thresholding at $T = 0.65$.

E. Training Protocol

HCC-Net was trained for 30 epochs using AdamW [14] with cosine annealing [15], weighted focal loss [16] ($\gamma = 2.0$), label smoothing ($\epsilon = 0.05$), and 5-fold patient-wise cross-validation. Full hyperparameters are listed in Table I.

HCC-RefineNet was trained using AdamW [14] with a hybrid Dice-Focal loss [16], [17] ($\alpha = 0.5$, $\gamma = 2.0$) and 5-fold patient-wise cross-validation. Model selection was based on validation Dice. Full hyperparameters are listed in Table II.

F. Evaluation Protocol

For HCC-Net, all experiments were conducted at the patient level using 5-fold patient-wise cross-validation to prevent data leakage. During training, the validation macro-F1 score was monitored after each epoch. Early stopping with a patience of six epochs was applied, and the checkpoint with the highest validation macro-F1 was retained for each fold. Performance

TABLE I: HCC-Net training configuration.

Training parameter	Value
Backbone	EfficientNet-V2-S
Input size	384×384
HU windowing	[0, 300]
Batch size	12
Epochs	30
Optimizer	AdamW
Learning rate / weight decay	$2 \times 10^{-4} / 1 \times 10^{-4}$
Scheduler	Cosine annealing with warm restarts ($T_0 = 10$)
Loss	Weighted focal ($\gamma = 2.0$) + label smoothing ($\epsilon = 0.05$)
Class weights	Inverse class frequency
Mixed precision	Enabled (AMP)
Early stopping	Patience = 6 (macro-F1)
Cross-validation	5-fold (patient-wise)

TABLE II: HCC-RefineNet training configuration.

Training parameter	Value
Voxel spacing	$1.5 \times 1.5 \times 1.5$ mm
HU windows	w1: [-100, 400], w2: [-160, 240], w3: [0, 200]
Smoothing	Gaussian ($\sigma = 0.75$)
ROI size (training patch)	$96 \times 96 \times 64$
ROI dilation kernel	$5 \times 5 \times 5$
Samples per volume	6
Optimizer	AdamW
Learning rate / weight decay	$1 \times 10^{-4} / 1 \times 10^{-5}$
Loss	Dice + Focal ($\alpha = 0.5$, $\gamma = 2.0$)
Max epochs	50
Batch size	2
Cross-validation	5-fold (patient-wise)

TABLE III: Comparison of backbone architectures for HCC/ICC/cHCC-iCCA classification. Best results are shown in **bold**.

Model	Accuracy	macro-F1	macro-AUC	F1(HCC)	F1(ICC)	F1(cHCC-iCCA)
ResNet-50	0.94	0.94	0.98	0.96	0.93	0.94
EfficientNet-B0	0.93	0.93	0.97	0.94	0.92	0.94
EfficientNet-B1	0.94	0.94	0.97	0.94	0.93	0.93
EfficientNet-V2-S	0.96	0.96	0.98	0.96	0.94	0.97

TABLE IV: Per-class diagnostic performance (EfficientNet-V2-S, OOF).

Class	Prec.	Sens.	Spec.	F1	N
HCC	0.93	0.98	0.96	0.96	94
ICC	0.95	0.93	0.97	0.94	99
cHCC-iCCA	0.98	0.95	0.99	0.97	85

was evaluated using accuracy, macro-F1, macro-averaged one-vs-rest (OvR) ROC-AUC, and balanced accuracy. Predictions from the validation fold of each cross-validation run were aggregated into a single out-of-fold (OOF) set, with each patient contributing exactly one prediction from a model that did not observe that patient during training. Ninety-five percent confidence intervals were estimated on OOF metrics using non-parametric bootstrap resampling (1,000 iterations) at the patient level. The random seed was fixed for reproducibility.

For HCC-RefineNet, tumor segmentation performance was evaluated using 5-fold patient-wise cross-validation, with folds stratified by the presence of tumor annotations. At inference, volumes were processed using the same preprocessing pipeline as training, without data augmentation. Sliding-window inference with 50% overlap was employed. Network logits were converted to probabilities via the sigmoid function, and a fixed threshold of 0.65 was applied to obtain binary tumor masks. This threshold was selected based on validation experiments evaluating thresholds in the range of 0.5 to 0.95. Segmentation performance was evaluated on the full 3D volume for each patient after ROI masking. From voxel-wise comparisons between predicted and ground-truth masks, Dice, IoU, Precision, Recall, Specificity, HD95, and MSD were computed. Predictions from all five folds were aggregated into a single OOF set, where each patient was evaluated only by the model that did not observe its data during training. OOF performance was reported as mean $\pm$ standard deviation across folds.

III. RESULTS

A. Classification Result (HCC-Net)

HCC-Net achieved an accuracy of 0.96, macro-F1 of 0.96, and macro-AUC of 0.98 under OOF evaluation, with balanced performance across all three subtypes (Table V). EfficientNet-V2-S outperformed ResNet-50 and EfficientNet-B0/B1 across most metrics (Table III). Per-class performance was consistently strong across all subtypes, with specificity at or above 0.96 for all classes (Table IV).

Phase ablation shows that combining all four phases (C1+C2+C3+P) yields the best performance, with the

TABLE V: Overall performance of the proposed model on OOF evaluation. Metrics include Accuracy, macro-F1, macro-AUC (OvR), and Balanced Accuracy with 95% confidence intervals.

Metric	Value	95% CI (Low)	95% CI (High)
Accuracy	0.96	0.93	0.98
macro-F1	0.96	0.93	0.98
macro-AUC (OvR)	0.98	0.97	0.99

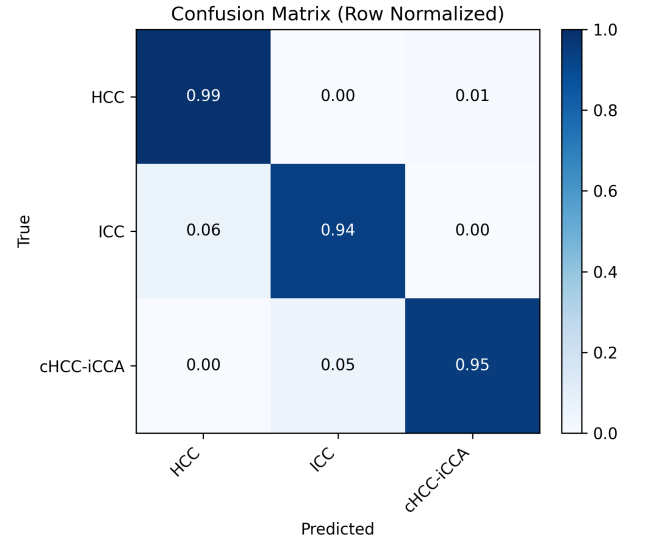


Fig. 3: Row-normalized confusion matrix for HCC-Net on OOF predictions.

plain phase contributing complementary information beyond contrast-only inputs (Table VI).

Figure 3 shows the row-normalized confusion matrix; most misclassifications occur between ICC and cHCC-iCCA, consistent with their overlapping enhancement patterns. ROC curves confirm high separability across all three subtypes (Figure 4). Precision-Recall curves are shown in Figure 5, further confirming consistent per-class discrimination. On the external WAW-TACE cohort (233 HCC cases), the 2D model achieved a sensitivity of 0.79, demonstrating reasonable cross-center generalization.

B. Segmentation Refinement Results (HCC-RefineNet)

HCC-RefineNet achieved consistent segmentation performance across phase configurations (Table VII). The three-phase configuration (C1+C2+C3) yielded the best boundary metrics

TABLE VI: Phase ablation study using EfficientNet-V2-S. Best results are shown in **bold**.

Phases	Accuracy	macro-F1	macro-AUC	F1(HCC)	F1(ICC)	F1(cHCC-iCCA)
P	0.94	0.94	0.98	0.93	0.92	0.97
C1	0.94	0.94	0.98	0.94	0.92	0.96
C2	0.95	0.95	0.97	0.94	0.95	0.95
C3	0.95	0.95	0.98	0.95	0.94	0.97
C1 + C2	0.94	0.95	0.97	0.94	0.94	0.95
C1 + C3	0.94	0.94	0.97	0.95	0.94	0.95
C2 + C3	0.95	0.95	0.98	0.96	0.94	0.95
C1 + C2 + C3	0.93	0.93	0.97	0.93	0.92	0.95
C1 + C2 + C3 + P	0.96	0.96	0.98	0.96	0.94	0.97

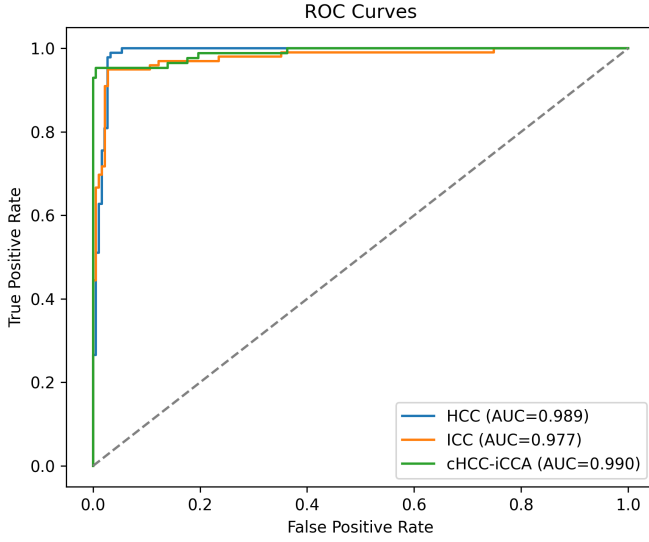


Fig. 4: ROC curves (OvR) for HCC-Net on OOF predictions.

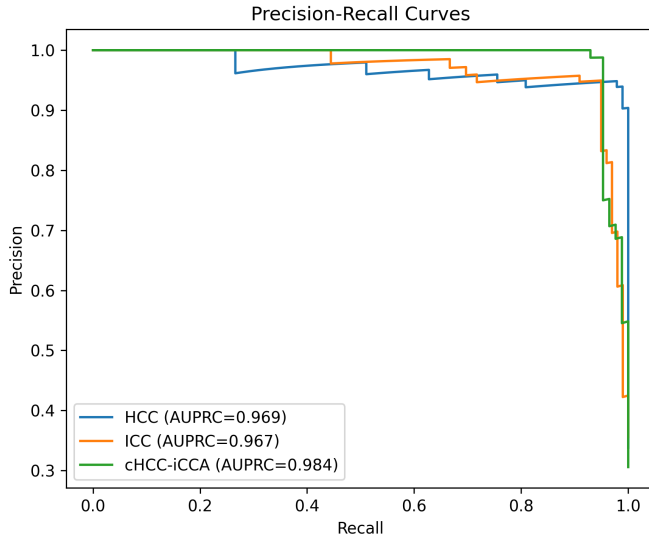


Fig. 5: Precision-Recall curves for HCC-Net on OOF predictions.

(HD95: 2.99, MSD: 0.86), while voxel-wise overlap metrics were similar across all configurations.

IV. DISCUSSION

This study evaluated whether multiphasic CECT can support three-way primary liver cancer subtype classification (HCC, ICC, cHCC-iCCA) and post-hoc tumor boundary refinement using two complementary deep learning models. Internal experiments show that high subtype discrimination performance can be achieved using an annotation-free 2D middle-slice representation with phase-stacked inputs, while accurate 3D tumor segmentation can be obtained with limited dependence on the specific phase configuration. For subtype classification, single-phase models exhibited robust internal performance. However, the integration of all four phases yielded the highest overall metrics. Notably, combining all three contrast phases (C1 + C2 + C3) without the plain phase yielded slightly lower performance than single-phase models, while the best results were achieved when all four phases were combined (C1 + C2 + C3 + P). Prior work has demonstrated that unenhanced CT encodes discriminative features for differentiating hepatic lesion types [18], and the ablation results confirm this for subtype classification, where adding the plain phase yielded the highest overall performance (Table VI). In contrast, for segmentation, tumor boundaries are primarily delineated by contrast uptake differences visible in enhanced phases [19], where the plain phase contributes minimal additional edge information. Performance varied only modestly across phase configurations. A C1-only model achieved high volumetric agreement, while adding C2 and C3 primarily affected boundary-based metrics, reflected by lower HD95 and MSD. Many prior studies focus on binary classification or tri-class settings that do not explicitly include cHCC-iCCA as a separate category, or rely on radiomics and clinical variables to achieve competitive performance [6]–[8]. Among studies reporting external validation, Huang et al. [6] evaluated their model on an external cohort of 83 cases. In contrast, the external test set in this study comprised 233 patients, providing a larger assessment of cross-center generalization. Several limitations should be acknowledged. Both models were trained on a single dataset, and external validation on WAW-TACE [11] included only HCC cases, precluding specificity assessment for ICC and cHCC-iCCA across sites. Classification relied on a middle-slice assumption that may miss off-

TABLE VII: Segmentation performance across different phase input configurations.

Phases	Dice	IoU	Precision	Recall	Specificity	HD95	MSD
C1	0.93 $\pm$ 0.05	0.87 $\pm$ 0.08	0.90 $\pm$ 0.09	0.96 $\pm$ 0.02	0.99 $\pm$ 0.00	3.05 $\pm$ 0.82	0.87 $\pm$ 0.30
C1 + C2	0.92 $\pm$ 0.05	0.86 $\pm$ 0.08	0.89 $\pm$ 0.10	0.96 $\pm$ 0.02	0.98 $\pm$ 0.00	3.17 $\pm$ 0.86	0.92 $\pm$ 0.30
C1 + C2 + C3	0.93 $\pm$ 0.05	0.87 $\pm$ 0.08	0.90 $\pm$ 0.09	0.96 $\pm$ 0.02	0.99 $\pm$ 0.00	2.99 $\pm$ 0.77	0.86 $\pm$ 0.28

center tumors, though 74% of patients had a tumor-containing slice within ± 5 slices of mid. Segmentation was evaluated within ground-truth-derived ROIs, likely overestimating fully automated performance, and no explainability method (e.g., Grad-CAM) was applied, nor direct comparison with recent state of the art methods. The three-phase configuration (C1 + C2 + C3) underperformed compared to several single-phase models (Table VI), possibly due to redundant signals without the plain-phase channel. Future work should prioritize multi-center validation, volumetric classification, and explainability integration.

V. CONCLUSION

This paper evaluated the use of multiphasic CECT for classifying liver cancer subtypes and refining post-hoc tumor boundaries using two complementary deep learning models. HCC-Net achieved high internal classification performance across HCC, ICC, and cHCC-iCCA and achieved a sensitivity of 0.79 on an external HCC-only cohort. Meanwhile, HCC-RefineNet produced consistent 3D segmentation performance across phase configurations, with improved boundary-based metrics when multiple contrast phases were used. Together, these results indicate that subtype classification and tumor mask refinement can be derived from the same multiphasic image data without reliance on additional clinical inputs.

Future work may include multi-center validation and evaluation under heterogeneous acquisition conditions.

ACKNOWLEDGMENT

This research was supported by The VNUHCM-University of Information Technology's Scientific Research Support Fund.

DATA AVAILABILITY STATEMENT

All datasets used in this study are publicly available. The multiphasic liver CT dataset is available via ScienceDB [10], and the external WAW-TACE dataset is available via Zenodo [11]. No proprietary data were used. The source code for this study is available at <https://github.com/SantiagoDeStefano/multiphasic-ct-liver-classification-segmentation-refinement>.

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